

Topics: 12.1 Three-Dimensional Coordinate Systems; 12.2 Vectors; 12.3 The Dot Product

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1. Consider the point $P(3, 4, 5)$.
 - (a) What is the projection of the point onto the xy -plane?
 - (b) What is the projection of the point onto the xz -plane?
 - (c) Find the length of the line segment $\overline{OP}$, where $O(0, 0, 0)$ is the origin.
 - (d) Find the position vector for the point $P(3, 4, 5)$ in **ijk**-form.
2. Find an equation of a sphere if one of its diameters has endpoints at $(1, 2, 4)$ and $(4, 3, 10)$.
3. Given $\mathbf{a} = 9\mathbf{i} - 8\mathbf{j} + 7\mathbf{k}$ and $\mathbf{b} = \langle 7, 0, -9 \rangle$, find the following. Simplify your answers completely.
 - (a) $\mathbf{a} + \mathbf{b}$
 - (b) $3\mathbf{a} - \mathbf{b}$
 - (c) $|\mathbf{b}|$
 - (d) $|\mathbf{b} - \mathbf{a}|$
4. Find the vector that has the opposite direction to $\langle 9, -6, -2 \rangle$ and has length 5.
5. Determine whether the given vectors are orthogonal, parallel, or neither.
 - (a) $\mathbf{a} = 4\mathbf{i} - \mathbf{j} + 4\mathbf{k}$ and $\mathbf{b} = 5\mathbf{i} + 12\mathbf{j} - 2\mathbf{k}$
 - (b) $\mathbf{a} = \langle 6, 5, -2 \rangle$ and $\mathbf{b} = \langle 5, 0, 9 \rangle$
 - (c) $\mathbf{a} = \langle -18, 15 \rangle$ and $\mathbf{b} = \langle 12, -10 \rangle$
6. Given vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ in V_3 , which of the following expressions are meaningful? Which are meaningless? Explain your reasoning.
 - (a) $(\mathbf{a} \cdot \mathbf{b}) \cdot \mathbf{c}$
 - (b) $(\mathbf{a} + \mathbf{b}) \cdot \mathbf{c}$
 - (c) $\mathbf{a} + (\mathbf{b} \cdot \mathbf{c})$
7. Find the angle θ between the vectors $\mathbf{a} = \langle 2, 0, -3 \rangle$ and $\mathbf{b} = -2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$.
8. Consider $\mathbf{a} = \langle -1, 4, 8 \rangle$ and $\mathbf{b} = \langle 18, 2, 1 \rangle$.
 - (a) Find the scalar projection of $\mathbf{b}$ onto $\mathbf{a}$.
 - (b) Find the vector projection of $\mathbf{b}$ onto $\mathbf{a}$.
9. Find the work (in joules) done by the constant force $\mathbf{F} = 8\mathbf{i} - 6\mathbf{j} + 5\mathbf{k}$ acting on an object that moves from the point $(0, 6, 4)$ to the point $(4, 14, 22)$ along a straight line. The distance is measured in meters and the force in newtons.

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Dot Product If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, then

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

If θ is the angle between nonzero vectors $\mathbf{a}$ and $\mathbf{b}$, then

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta.$$

Scalar and Vector Projections If $\mathbf{a} \neq \mathbf{0}$, then the *scalar projection* of $\mathbf{b}$ onto $\mathbf{a}$ is

$$\text{comp}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|}.$$

The *vector projection* of $\mathbf{b}$ onto $\mathbf{a}$ is

$$\text{proj}_{\mathbf{a}} \mathbf{b} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|} \right) \frac{\mathbf{a}}{|\mathbf{a}|} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|^2} \right) \mathbf{a}.$$

Work by a Constant Force If a constant force $\mathbf{F}$ moves an object through displacement $\mathbf{D}$, then the work done is

$$W = \mathbf{F} \cdot \mathbf{D}.$$

Suggested Textbook Problems

Section 12.1: 1-46

Section 12.2: 1-33, 37-44

Section 12.3: 1-56